Forget the Recording?
Open Speech Models Remember Almost as Well from a Transcript

Hanyu Yang

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Abstract

A speech assistant with long-term memory can keep a user's old recordings or only a written record of them. On acoustic questions from VoxMem 8K, two open readers gain nothing reliable from the recordings. Qwen3-Omni scores 19.0 points higher with the full audio history than with no memory, yet only 1.4 points higher than with a plain transcript (95% interval −5.4 to +7.3; 142 held-out question families). For MiniCPM-o 4.5 the transcript is non-inferior to the audio at a preregistered 10-point margin, but only narrowly. On the same held-out families, trading part of a labelled text record for old clips at the same size in MiniCPM-o context positions moves accuracy by +2.8 points for MiniCPM-o and +2.1 for Qwen3-Omni; both intervals include zero. Gemini 3.8 Flash showed an audio gain in an exploratory run whose sealed rerun did not finish. Stored as uncompressed audio, the recordings take about 300 times the bytes of the labelled text.

1 Introduction

A speech assistant that remembers last week's conversation has to decide what to keep: your recordings, or a written note of them. We ask which one a frozen speech language model answers better from when the question is about the sound itself. Who spoke? In what mood? Over what background noise?

The choice matters because audio is heavy. A minute of speech fills about 600 context positions in MiniCPM-o, and as uncompressed audio its file is hundreds of times larger than a text note.

Xiao, Sethu and Holden built VoxMem to test this kind of memory [1]. In their check that the questions need the audio, Gemini 3.7 Flash falls from 46.2% with audio to 4.4% with the script the speech was synthesized from.

A transcript drops the voice, so we write the voice down too. Each user turn gets speaker, emotion and sound-event labels next to its words. Then we hold the size fixed in MiniCPM-o positions and trade part of this labelled text for old clips. If the audio carries something the labels miss, the trade should pay. For both open readers it does not pay reliably. Qwen3-Omni does use its memory: the full audio history lifts it 19 points over no memory. Yet a plain transcript gets it to within 1.4 points of that.

We run three preregistrations: MiniCPM-o 4.5 on 300 test and 142 held-out question families, then Qwen3-Omni on the same holdout. Gemini 3.8 Flash gained from audio in an exploratory run, and that result stays open.

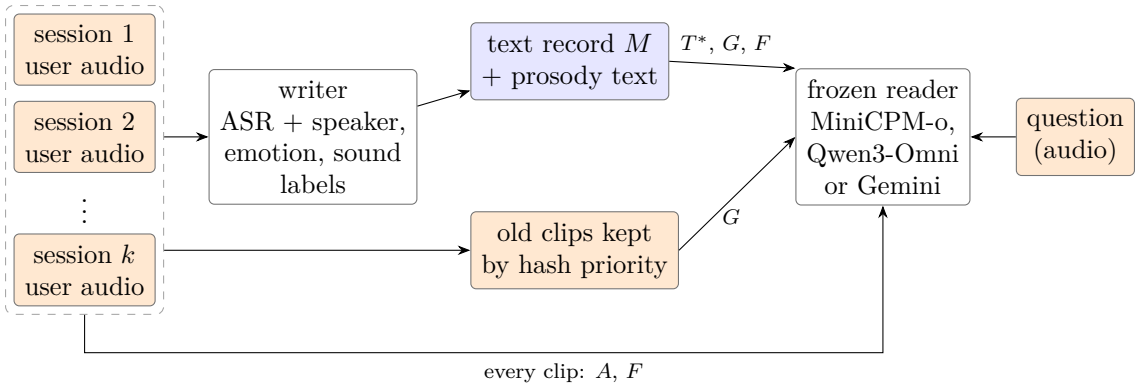


Figure 1: Memory construction for T^*, G, F and A. The writer processes each session once; the reader receives one memory and the spoken question.

2 What we did

Each VoxMem item is a synthesized multi-session conversation at the 8K length, followed by a spoken question about an earlier turn. We use the answerable acoustic questions and split them by question family. No session is evidence in two splits, but sessions recur across splits as filler, some of them evidence elsewhere.

A writer reads each session once, in order (Fig. 1). For every user turn it stores its own transcript and key–value labels; together these form the text record M. The budget is $B_f = P_M + \lceil f P_A \rceil$, where P_M is the size of M in MiniCPM-o positions, P_A the size of the user audio, and $f = 0.25$. The memory arms are:

- N: no memory, only the question.
- M_{asr}: the transcript alone, without labels (holdout only).
- A: the full audio history, the VoxMem setting.
- T^*: M plus per-turn prosody numbers, filled to B_f.
- G: M plus old user clips kept online by hash-priority eviction, then prosody numbers up to the same B_f, so it holds fewer of them than T^*. Audio is about 9% of it (Fig. 2).
- F: T^* plus every user clip, over budget.

Every history arm includes the assistant's text turns. Every memory is re-read from scratch. Only G and T^* are matched in size; A and M_{asr} have different costs.

The writer uses SenseVoice for ASR [2], emotion2vec for emotion [3], AST for sound labels [4] and WavLM for speaker features [5]. Speaker labels are local to each session. Settings were frozen on 32 dev families.

MiniCPM-o's primary test was a gate: F must significantly beat T^* before we claim an audio benefit. If it does not, $G - T^*$ is reported as an estimate only. On the holdout, the primary test asks whether M_{asr} is at most 10 points worse than A, a one-sided non-inferiority test.

A first judge scores answers blind to the arm: Gemini 3.8 Flash for MiniCPM-o, Claude Opus 5.5 for Gemini and Qwen. A second judge sees a seeded 10% plus every first-judge/string-matcher disagreement. For Qwen this is Claude Sonnet 5.5, agreeing on 121 of 124 distinct answers (204 of

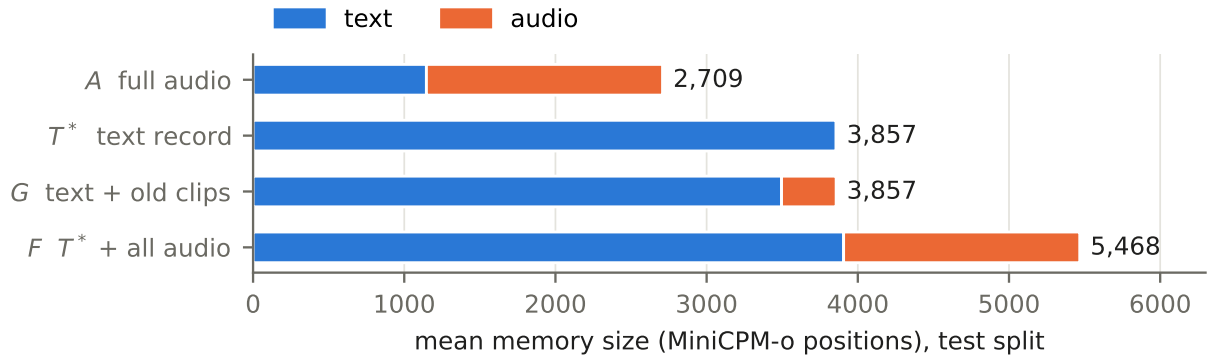


Figure 2: Mean memory size per arm on the test split, divided into text and audio positions. T^* and G have the same size for every family, to within one position. Text in A is session headers, turn markers and the assistant's turns.

208 rows). The planned tag-word canonicalisation and owner adjudication were not done. Intervals are type-stratified cluster bootstrap-t over 48 test units and 35 holdout units (20 for Gemini's subset); each unit takes the evidence type of most of its families. One-sided p-values come from a cluster sign-flip test, so the two can disagree near a boundary. Clustering by every shared voice or background clip merged nearly everything; we use the preregistered fallback of one dominant source per family.

Qwen3-Omni-30B-A3B-Instruct uses its thinker with greedy decoding and at most 48 new tokens. It completed 142 families across $N, A, M_{\text{asr}}, T^*, G$: 710 rows without errors. Its inputs exclude gold answers. Qwen and Gemini reuse memories counted in MiniCPM-o positions; Qwen's native input lengths differ. Gemini accesses clips through the `agy` agent CLI.

3 What we found

MiniCPM-o: no reliable benefit from added clips. $F - T^* = -0.3$ points (-5.4 to $+5.7$), so the gate is not passed and $G - T^*$ is an estimate only: $+0.7$ (-3.2 to $+5.6$) on test. On holdout it is $+2.8$ (-1.2 to $+7.1$), or $+4.2$ ($+0.4$ to $+8.3$) with the second judge's verdicts where it saw the answer.

MiniCPM-o, holdout: a bare transcript is non-inferior to full audio. $A - M_{\text{asr}} = -1.4$ (-10.1 to $+9.3$). The upper bound clears the 10-point margin by 0.7 points; an unstratified bootstrap over the same units gives $+10.4$, which misses it. Separately, adding one label group at a time to the transcript passed none of its tests.

Qwen3-Omni: no reliable gain over a transcript. $A - N = +19.0$ points (10.8 to 27.7). Against a transcript, $A - M_{\text{asr}} = +1.4$ (-5.4 to $+7.3$; one-sided $p = 0.42$), giving the preregistered decision NOT-SHOWN. It is 0.0 (-6.8 to $+6.7$) under the matcher and $+0.7$ (-5.9 to $+6.8$) with the second judge's verdicts. The reader uses its memory. The audio's extra value over text remains uncertain. $A - T^* = +5.6$ (-3.3 to $+13.6$), so labelled text fails to establish equivalence within 10 points. $G - T^* = +2.1$ (-0.9 to $+6.2$) at the MiniCPM-o budget; in Qwen's own positions G is a median 104 longer than T^*, which favours G. Against MiniCPM-o's -1.4 on the same families, the audio-over-transcript contrast differs by $+2.8$ (-6.1 to $+11.9$) with different first judges, or $+2.8$ (-4.7 to $+10.8$) under the shared matcher.

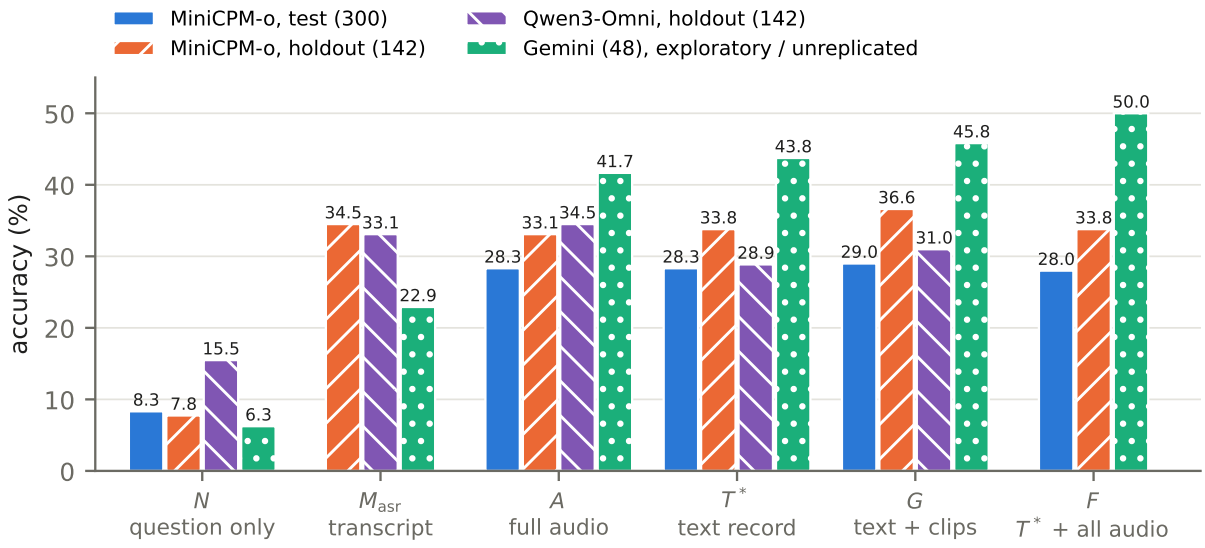


Figure 3: Accuracy under each reader's first judge; parentheses give family counts. Gemini covers 48 holdout families and remains exploratory and unreplicated. Missing bars denote untested arms.

Gemini: exploratory and unreplicated. On 48 families, $A - M_{\text{asr}} = +18.8$ (6.6 to 31.7); T^* scores 43.8% against 41.7% for A. The agent could access answer files. Two shell-using transcript runs remained and were scored wrong; counting their answers gives +14.6 (1.5 to 27.0; $p = 0.057$; post hoc). A sealed rerun stopped at 324 of 710 rows on quota, unscored.

Storage and context cost differ. On MiniCPM-o's test split, the estimated median storage for T^* is about 1/300 of A (4 bytes per text position, uncompressed 16-bit audio). Yet the median T^* memory uses about 45% more context positions than A.

4 Where this stops

We did not test a retained key–value cache. The freeze record comes from a private repository and is self-attested. No contrast shows that either open reader uses old clips at all, so "audio adds nothing" cannot be told apart from "these readers cannot use old audio". The resampling units come from one dominant source per family; families in different units can still share a voice or clip, which can make the intervals too narrow. On the holdout, voice-cluster purity was 0.89 against a 0.9 target and clip-match accuracy 0.58 on 19 checked clips. For the open readers the judge and the matcher disagree most on A, up to 6 points more than on the text arms; the preregistration treats a gap above 5 points as a threat. The strict matcher puts MiniCPM-o's full-audio test accuracy at 4.7%, showing that scoring matters. These results concern synthesized speech at one history length; reader, writer and benchmark effects remain entangled.

Would Gemini still gain from audio with sealed inputs and no access to answer files? The unfinished rerun leaves that question open.

Data and code. The public repository contains the code, preregistrations and outputs.

AI assistance. AI models assisted with drafting the prose and figures.

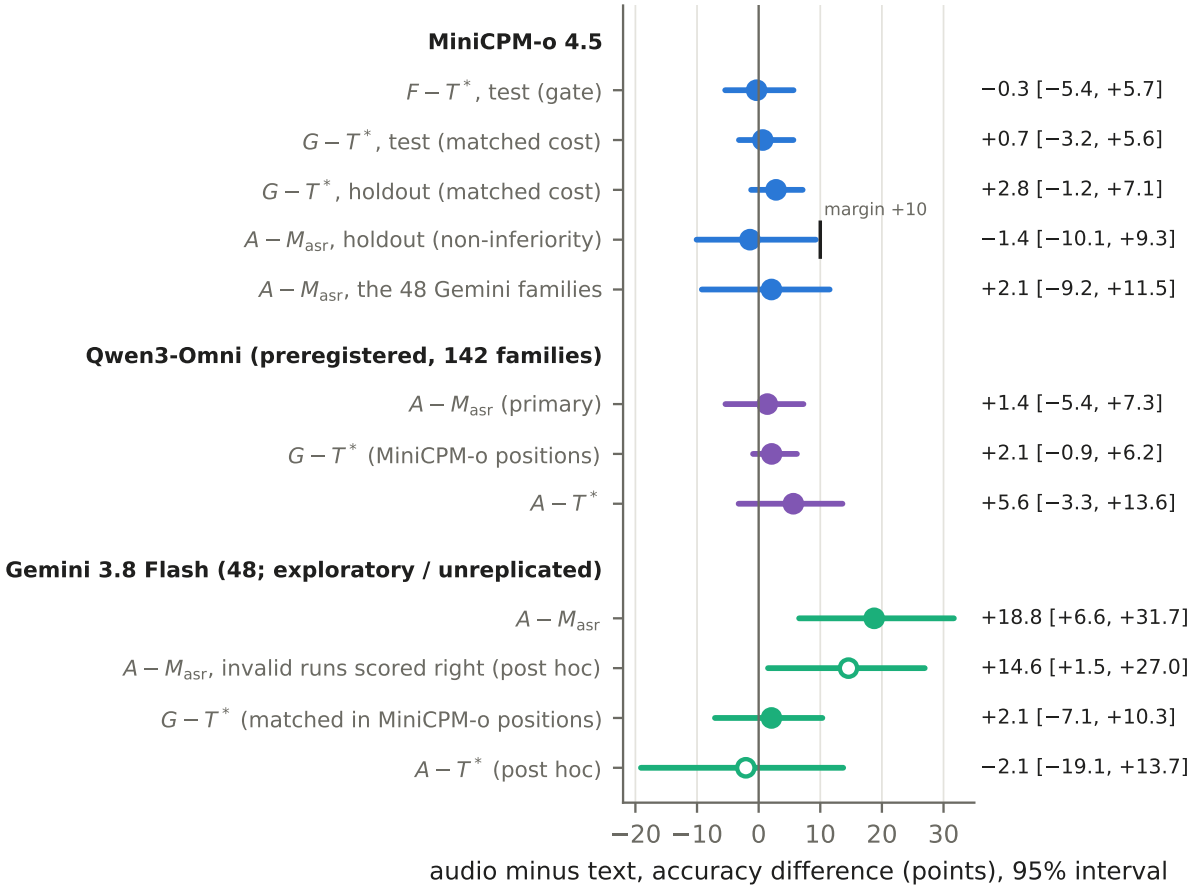


Figure 4: Paired differences, each oriented as audio minus text, so points to the right favour audio. Bars are 95% type-stratified cluster-bootstrap intervals; transcript non-inferiority requires the upper endpoint below the +10 tick. Open circles are post hoc.

References

[1] Y. Xiao, V. Sethu, E.-J. Holden et al. VoxMem: Benchmarking multimodal memory in large audio language models. arXiv:2609.32607, 2026.

[2] K. An, Q. Chen, C. Deng et al. FunAudioLLM: Voice understanding and generation foundation models for natural interaction between humans and LLMs. arXiv:2407.04051, 2024.

[3] Z. Ma, Z. Zheng, J. Ye et al. emotion2vec: Self-supervised pre-training for speech emotion representation. arXiv:2312.15185, 2023.

[4] Y. Gong, Y.-A. Chung, J. Glass. AST: Audio spectrogram transformer. arXiv:2104.01778, 2021.

[5] S. Chen, C. Wang, Z. Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. arXiv:2110.13900, 2021.